

Network Measurement as a Cooperative Enterprise

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Abstract. Real-time network measurements can be used to improve performance of existing Internet services and support the deployment of new services dependent on performance information (e.g., topologically-aware overlay networks). Internet-wide measurement faces numerous scaling-related challenges, including the problem of deploying enough measurement endpoints for wide-spread coverage. We observe that peer-to-peer networks, made up of “volunteer” hosts around the Internet world, have the potential to provide a level of coverage that greatly exceeds that made possible with the tedious human process of negotiating endpoint locations. We therefore propose a distributed peer-to-peer system that can be queried for network performance information. We sketch the architecture and operation of such a system and briefly relate it to alternative proposals for measurement infrastructures. Finally, we list open problems related to the design and realization of such a system.

1 Introduction

Measurements of network performance are valuable for improving performance, assessing utilization, engineering traffic and validating design choices. We are particularly interested in real-time measurements that can be used to improve the performance of existing Internet services and support the deployment of new services dependent on performance information (e.g., topologically-aware overlay networks).

The challenges involved in constructing an Internet-scale measurement infrastructure are considerable. First, there is the difficulty of coverage, that is, obtaining access to a large number of distributed measurement endpoints. Current measurement systems generally involve human-negotiated access to endpoints either with ISPs and/or friends at diverse locations [3, 6]. Second, there is the difficulty of obtaining accurate measurements, given the time-varying nature of network properties of interest (e.g., loss rate, available bandwidth, latency). Third, there is the issue of overhead. Care must be taken to avoid a measurement process that imposes excessive overhead on the overall system. These challenges

have obvious interactions; for example, one can reduce overhead with less accurate measurements or more coarse-grained coverage.

We observe that peer-to-peer networks, made up of “volunteer” hosts around the Internet world, have the potential to provide a level of coverage that greatly exceeds that made possible with the tedious human process of negotiating end-point locations. *We therefore propose a distributed peer-to-peer system that can be queried for network performance information.* The M-coop (or Measurement co-operative) is a system that answers queries about the path between two arbitrary IP addresses. In addition to performance metric information, the system returns assessments of the metric accuracy and trustworthiness.

Such a system does not, on its own, solve the problem of obtaining accurate measurements. Nor does it solve the problem of measurement overhead. Indeed, because such a system may involve a very large number of end systems, the scaling problem is significant. We will rely on known techniques for dealing with accuracy (e.g., using moving weighted averages); we will introduce mechanisms for reducing the number of end systems that form measurement pairs to help with the scale problem.

Such a system brings with it a number of additional challenges. Well-known are the problems that result from a peer-to-peer system of hosts that may join and leave on a frequent basis [4, 5, 8, 9]. Merely keeping the M-coop system connected can be challenging in this environment. Because the measurement entities are volunteers, and not under any accountable control, we must deal with issues of inaccurate information due to misconfiguration or malicious use. The inclusion of a trustworthiness value recognizes the fact that information quality may vary. We must also consider the question of incentive. What would motivate someone to include their host in an M-coop measurement infrastructure? The limited examples of deployed peer-to-peer systems indicate that people are motivated by self-interest (e.g., Napster, Gnutella) and by a sense of contributing to a larger “good” (e.g., SETI@home). An Internet-scale measurement infrastructure has the potential to tap both sources of motivation.

In the next section, we sketch the design of an M-coop system. In Section 3, we describe some details of the architecture and briefly sketch the operation of this system. In Section 4, we briefly describe related work and then conclude with a section discussing the open problems in the design and realization of an Internet-wide peer-to-peer measurement system.

2 An M-coop Design

We sketch one possible design of a cooperative measurement system. The system has some features in common with other measurement infrastructures (most notably IDMaps [1] and NIMI [3]). Some similarities and differences are discussed in the Related Work section.

2.1 The Service

The M-coop system answers queries of the form $(IP1, IP2, measurement\ type)$ where $IP1$ and $IP2$ are IP addresses. The measurement type may be any network quantity measurable by hosts on a network, e.g., delay, bandwidth, jitter. The system returns the answer to the query along with trust and accuracy parameters if available. As a voluntary peer-to-peer system, the possibility of misinformation is high, so a trust value is reported with the information returned. The trust value is an indication of the past reliability (with respect to quality of information) of the node that responded to the query.

The size of the Internet dictates that any measurement will only be an estimate. To keep the system manageable, instead of the measurement being from the requested host, it might be from a “nearby” node on the overlay network. Also, the measurement process may contain some inaccuracies due to changing paths in the Internet, inherent inaccuracies in the measurement process, congestion, etc. The accuracy value tries to quantify the “nearness” of the host to the measurement node as well as the inaccuracies in the measurement process.

We do not address the question of who is allowed to make queries. In the spirit of cooperatives, one might imagine that only participants are allowed access to the community information. This sort of access control (or any other) is orthogonal to the base system design.

2.2 The Architecture

Architecturally, the system consists of two overlay networks, a logical overlay formed as a Distributed Hash Table (DHT), (e.g. Chord [8], SCAN [4], Pastry [5] and Tapestry [9]), and a measurement overlay (m-overlay) based on the AS level graph of the Internet. Each node in M-Coop belongs to both overlays. The DHT is used for storing and looking up information about the nodes that compose the overlays. Nodes connected by edges in the m-overlay form measurement peers. The key of a node in the DHT is generated by its neighbours during the join process while the key of a node in the m-overlay is its IP address. An important issue is the construction of the m-overlay graph to support accurate measurements without undue overhead. Measurements are taken by the end-points of the edge in two ways, actively, by sending probe packets to each other, and passively, by monitoring the system traffic that traverses this edge of the graph.

For scalability, each node on the network is assigned an “area of responsibility” (AOR), defining a set of addresses for which it can answer queries. The AOR is assigned when the node joins the network. It changes as other nodes join and leave the network.

A query to the system, $(IP1, IP2, measurement\ type)$ is first routed through the DHT to the node which has $IP1$ in its AOR. We denote this node $R(IP1)$ to indicate it is responsible for $IP1$. If the measurement information is available, node $R(IP1)$ will reply, along with the available accuracy and trust information. If the data is not available, it may trigger a measurement or a new query. This

new query, called a composition query, will traverse a path on the m-overlay from $R(IP1)$ to $R(IP2)$ collecting metric data about the links traversed. This data is then returned as the reply to the composition query and finally as a reply to the original query.

A node on the system thus consists of three modules:

1. **Routing.** This module is responsible for maintaining the overlays, communicating with the peer nodes and routing queries and responses through the overlays.
2. **Measurement.** This module performs measurements between itself and its measurement peers, verifies the measurements obtained by other nodes and responds to queries about the node's AOR.
3. **Trust.** This module maintains the trust database, performs trust metric calculations and responds to trust queries.

The next section describes the architecture in more detail.

3 Architecture Details and Operation

3.1 Routing

There are many ongoing research efforts trying to develop better methods of locating data in a distributed system. These efforts are directed towards scalability, reliability, graceful degradation under dynamic conditions, and efficient search [4, 5, 8, 9]. Based on this, we assume that a method of locating data in the DHT is available to us and we will use the generic term *routing* to imply that a packet is using one of the above methods to reach its destination node.

The routing module is responsible for routing data queries and their responses through the DHT overlay based on the AOR of the nodes. Composition queries are routed in the m-overlay. Ideally, the routing on this overlay is the same as the network-layer routing, but this requires knowledge of BGP policies of each AS. To approximate this in the application-layer, a heuristic such as shortest-AS path may be used.

3.2 Measurement

Measurements are taken from the node to its measurement peers. The set of measurement peers is identified when the node joins the system and is updated as needed when peer nodes join or depart. The set of measurement peers is selected to map onto the underlying network topology in the following fashion. If a node in the overlay network is the only one in its autonomous system (AS), then it has one edge for each AS-level neighbor in the underlying network. If there are multiple nodes in the same AS, a clustering protocol is run when new nodes join or depart to ensure that there is only one node which has edges to the AS level neighbors. The remaining nodes are organized to provide redundancy and intra-AS measurement. The AS-level edges join this node to the m-overlay network

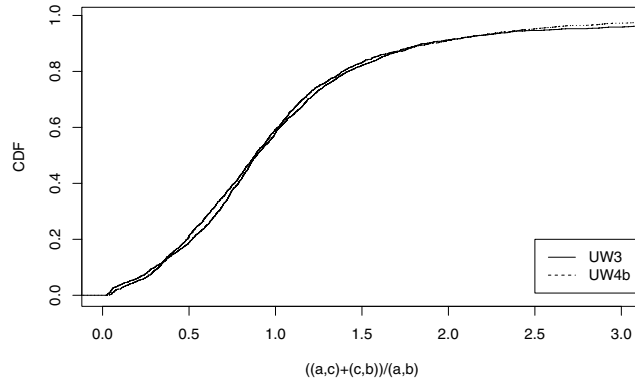


Fig. 1. CDF of composition of AS-level paths

nodes responsible for the IP addresses in the AS level neighbors. The intent is that the measurements obtained by the M-coop system will then better approximate the values seen by a packet on the underlying network. The measurement data obtained is stored with meta-information such as time of measurement, whether it is composed from other measurement data, whether it is cached, the accuracy, trust value, etc. Measurements may be taken periodically, triggered by options in a data query, and/or determined passively by examining packets in the system.

We performed some simple experiments to verify that such an AS-level composition of paths gives reasonable estimates of the original query between two IP addresses. We used data from the UW-3 and UW-4b datasets used for the Detour study[6]. These are end-to-end `traceroute` measurements collected from public `traceroute` servers. Details of the data collection method can be found in the referenced study. For each dataset, we calculate the average latency between two nodes a and b and also compute the AS-level path between them. The composition path is computed by finding a node c which lies on the AS-level path from a to b , and calculating $(a,c)+(c,b)$ for all possible such c 's. This simulates our scheme in which the composition packets are forwarded at the AS-level from the source AOR to the destination AOR. The results are shown in Fig. 1 as a CDF of the ratio $((a,c)+(c,b))/(a,b)$. It can be observed that most of the composition values are within a factor of two of the actual path value. These results are similar to those obtained by IDMaps [1] and show that forwarding at the relatively coarse level of ASes can give reasonable estimates.

3.3 Trust

Measurements are of two types, data and verification. Data measurements are performed between a node and its measurement peers and these are reported in response to queries. Verification measurements are performed to verify the

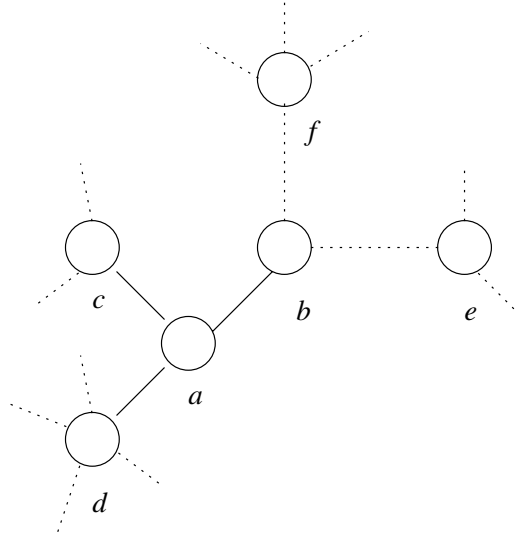


Fig. 2. Verification

responses of the peer nodes. These measurements are part of the process by which the trust module calculates the trust value to be assigned to the peer nodes.

The trust value of a node in the context of a particular link is a measure of its measurement reliability in the past. We assume that past behaviour is an indication of future actions, i.e., if a node has been providing reasonably accurate responses about a link in the past, it is likely to provide a reasonably accurate response when queried now. We will use the term “trust of a node” to implicitly mean the trust of a node with respect to a specific link.

The node checks the operation of its peers in the m-overlay using a *verification process*, which is run regularly. The results of this process are used to calculate the trust value of a node. This trust factors in the time since the last verification process was run, as well as reports from other nodes on the trust of the node in question. In the system, a node reports on the trust of its immediate neighbours. It also gathers information about nodes two and three hops away, which is then reported only if a query about that node’s trust passes through.

We now explain the verification process in more detail. In Fig. 2, node *a* responds to trust queries about the nodes *b*, *c* and *d*. Periodically, *a* runs the verification process on the nodes *b*, *c* and *d*. The verification process is in two parts: *a* queries *b* about the path from *b* to *e*, which is *b*’s neighbour; *a* also performs a measurement from itself to *e* directly. Since *a* knows the value of the measurement of the *a-b* link, it can estimate the *b-e* link and compare it with the value reported by *b*. The two values thus obtained are then used to update the trust of node *b*.

It is important to note that the b - e link measurement estimated by a has higher chances of being inaccurate and so a single value which doesn't tally with the reported value may not be enough to affect the trust value of the node b .

This verification mechanism requires that the point-to-point measurements made by the node be independent of the trustworthiness of the other end point, and hence reliable. This can be partly achieved if at least some of the measurements can be performed without the cooperation of the other node (perhaps at the operating system level). For example, a ping query to measure latency does not reach the application layer of the other end-point and hence is harder to affect maliciously. Given this assumption, a can verify the functioning of b by making a direct measurement to e to estimate the b - e link. Since b is in a 's neighbouring AS and similarly, e in b 's, it is reasonable to suppose that the direct measurement by a will produce a good estimate. Another solution could be to have a measure directly to other nodes in b 's AS, if they exist. These solutions partly address the problem of verification, but do not provide a completely reliable method of verifying a neighboring node's measurements.

Trust Computation Procedure The verification process is used as a building block to compute the trust value of a response to a query using a trust computation procedure. When a query is answered by a node, the querying node has no knowledge of the reliability of the reported result. To determine the trustworthiness of the result, the querying node must ask the neighbours of the responding node about its trustworthiness. To find out the trustworthiness of the neighbours, the node must query their neighbours, and so on. This recursive chain is terminated at the querying node itself. Effectively, the node is building a chain of trust from itself to the responding node. This chain follows the AS-level path from the querying node to the neighbours of the responding node on the m-overlay. Each node on the chain reports the trust it has on the node preceding it (which is its neighbour).

The actual procedure is as follows. A querying node, on receiving a response to its query, can decide to calculate the trustworthiness of the response. It looks up the node information of the responding node on the DHT and gets the list of neighbours. It then selects a set of nodes (potentially all) to send the trust query to. Each query is routed through the m-overlay to its destination (a neighbour of the responding node). The destination node replies with the trust it has on the responding node. As this reply makes its way back to the querying node, each node along the path adds the trust it has on the node from which it received the reply. At the querying node, it applies a function all the information to obtain the trust value of the responding node.

3.4 AOR Assignment and Overlay Construction

The AOR assignment for a node takes place when the node joins the overlay network. The startup procedure for nodes joining the network assumes two things:

a node is capable of finding its AS number;¹ and a node knows the IP address of an existing node on the overlay. (The case of the first node in the overlay is discussed separately.) A further assumption that is useful, but not required, is that a node has access to the list of ASes connected to its AS.² The creation and maintenance of the measurement overlay is performed by the following algorithms.

- **Join.** This is executed by a node joining the measurement overlay.
- **Failure recovery.** This is executed by the neighbours of a node that has failed.
- **Maintenance.** This is executed with low frequency to redistribute the AORs across the peers.

The invariant that these algorithms maintain, is that every AS is in the AOR of some node. When a new node joins the overlay, it may be assigned a part of the AS space as its AOR by splitting off portions of a small set of pre-existing nodes' AORs. When a node leaves the overlay, its AOR is merged into the AOR of a neighbour of that node, thus maintaining the invariant. In the following discussion, we restrict our attention to joins and departures in the m-overlay. We do not discuss the DHT join and leave algorithms as they are handled by the DHT schemes.

Join Procedure In Fig. 3, node n is a new node that is attempting to join the network. On startup, the node n contacts a node s that is already a member of M-Coop. The address of the node s is assumed to be known to node n through some out-of-band mechanism such as from a website or an ftp server. In its initial message, n advertises its entire AS as its AOR. Since the overlay already exists, some node b already has this AS in its AOR. Node s queries the DHT to find that node b and returns its address to n . Node n then contacts b with the same advertisement. There are two cases: b and n are in different ASes, or they are in the same AS. We shall consider the cases separately. When the nodes b and n are not in the same AS, n performs the following procedure:

- n uses its list of neighbouring ASes to claim them from the AOR of b . If the ASes appear in b 's list of neighbouring ASes as well, then they are assigned to n probabilistically.
- After the basic AOR for n is created, n looks up the nodes responsible for the ASes that are in its neighbour list but not in its AOR or in the AOR of b . These nodes are sent peer requests by n to make them peers in the m-overlay.
- If these nodes have b as a peer, they replace b by n , otherwise they add n as a peer.
- n also peers with b .

¹ A repository of AS information is available at www.arin.net/whois.

² NLANR maintains such a list at <http://moat.nlanr.net/AS/>.

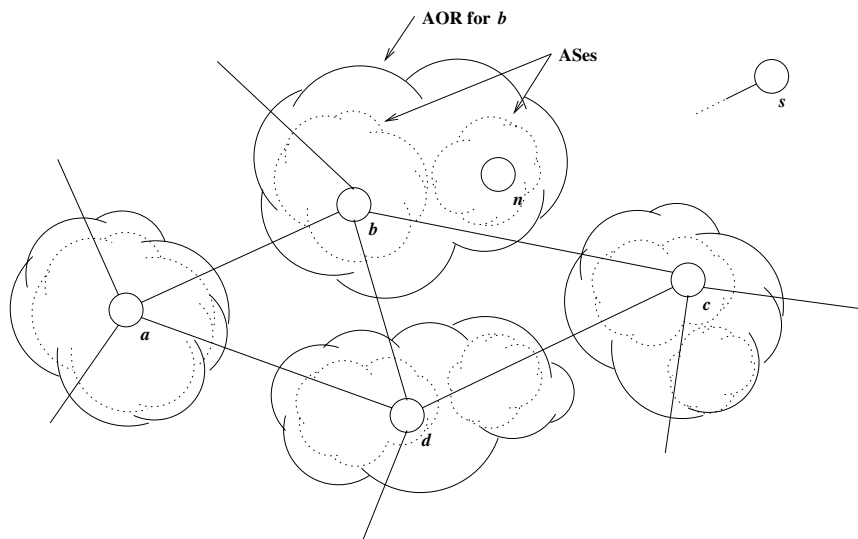


Fig. 3. Overlay before node n joins the network

- n runs the maintenance procedure to resize its AOR and potentially find new peers.

At this point, n is ready to begin making measurements with its peers. A key is generated for n by its peers and n uses this key to join the DHT and publish its peer and AOR information on the DHT. This marks the end of the join algorithm for n .

When the nodes b and n are in the same AS, n joins the cluster of nodes under b for that AS. The process is similar to that of the previous case except that the AORs contain IP prefixes, and the splitting of AORs is based on partitioning the set of IP addresses assigned to each node. Also, the node information of n is published in the intra-AS DHT.

When n is the first node in the overlay, it assigns all AS numbers to its AOR and waits for further nodes to join the network.

Note that the number of nodes that can join the measurement overlay at the intra-AS level is limited. This is to ensure that the AORs remain of reasonable size. Any nodes that join after this limit is reached do not participate in the measurement overlay but remain connected in the DHT. The information published in the DHT is soft state and must be refreshed periodically using the node key generated during the join process.

Failure Recovery The main elements for the failure recovery algorithm, namely the node information and the node key, are put in place during the join process. When a node fails or leaves the M-Coop network, it is the responsibility of its

neighbours in the m-overlay to recover from the departure. The periodic measurements taken between the peers in this overlay serve as a heartbeat mechanism and enable the detection of a departure. When a departure is detected, each neighbour of the missing node looks up the missing node information by querying the DHT using the missing node's node key. The neighbour also looks up the node information of the other neighbours to find the neighbour with the smallest AOR. This node incorporates the departed node's AOR into its own AOR, then peers with those nodes from the departed node's peers which are not already peers, runs the maintenance algorithm and then updates the node information published on the DHT.

Maintenance The objective of the maintenance procedure is to distribute the AORs across the peers so that each AS belongs in the AOR of the node closest to it. The maintenance algorithm consists of contacting each of the peers in the m-overlay in turn to negotiate the transfer of ASes based on the contents of the lists of neighbouring ASes. ASes are transferred only if necessary, i.e., if they are in a node's AOR but belong in the peer's neighbour list. This is to minimize the updates of the node information published in the DHT.

We are currently in the process of evaluating these algorithms using simulations.

4 Related Work

There have been several prior projects that concern measurement infrastructures (e.g., IDMaps [1], NIMI [3], SPAND [7] and Remos [2]). Our work is most closely to the IDMaps project, so we limit our related work discussion to that project.

IDMaps [1] is a proposal for a global infrastructure for gathering and distributing Internet host distance information. The goal of the IDMaps project is to provide distance metrics between two hosts on the Internet in an accurate and timely manner. The IDMaps architecture consists of a network of Tracers, which gather Internet distance information, and Clients, which use this information to estimate distances between hosts. The distance estimate between any two IP addresses is calculated from the Address Prefixes (APs) that contain the IP addresses, serving a similar function to our Areas of Responsibility. The calculation is performed by finding the APs to which the IP addresses belong, locating the systems or "boxes" to which the APs are closest and then running a spanning-tree algorithm to find the shortest distance between the two boxes. This calculation requires that a substantial portion of the box connection topology must be maintained.

The actual box-box topology can be achieved in two ways, the Hop-by-Hop (HbH) and the End-to-End (E2E) models. In the HbH model, every transit backbone router is modeled as a box and the calculation is the sum of inter-AS and intra-AS paths from one AP to the other. The distances on these paths are calculated by the Tracers probing the routers at random intervals. In the E2E

model, the Tracers are the boxes and the distances are calculated as the sum of the AP to box distances and the distance between the two boxes.

Our goals for the M-coop system are to provide a generalized metric collection and distribution infrastructure that is simple and rapid to deploy on a large scale. The information returned by the system also contains some indication of how reliable (in terms of accuracy and trustworthiness) the information is. Our approach to the problem is similar to the HbH model proposed in the IDMaps architecture but our method of distance estimation and information dissemination is fundamentally different. We intend to have a little more complexity at the nodes gathering the distance information to avoid the problem of maintaining a global view of the box topology. We also try to address the deployment of the system in the Internet by means of our peer-to-peer design.

5 Open Questions

We have sketched out the design and architecture of an Internet-wide measurement service. Some of the challenges that we are currently working on include:

- **Participation.** Will such a scheme generate enough participation to achieve critical mass, i.e., a level where the query results are a good approximation of the actual values? Related to this issue is the broader question of what will motivate people to participate in peer-to-peer systems. Will people eventually subscribe to peer-to-peer systems, like they subscribe to magazines? Or will they contribute their host to peer-to-peer systems, a la charitable donations? What are the best analogies to the peer-to-peer experience?
- **Generality.** Can a single system be used to satisfy the different measurement requirements of the diverse applications which might want to take advantage of this service? Can such a system be used as a common measurement service for peer-to-peer systems to use for optimizing their operation?
- **Usefulness of Parameters.** Can trust and accuracy be made useful to applications?
- **Composition.** Can composition of measurements from intermediate hops give meaningful values for the actual measurement between two IP addresses?
- **Collusion.** Collusion is a problem in trust systems. Can the amount of collusion required to subvert the system be made large enough to deter attacks?

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